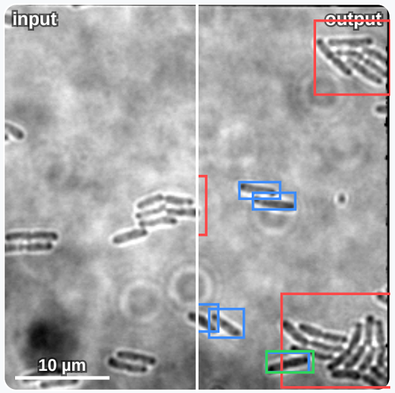
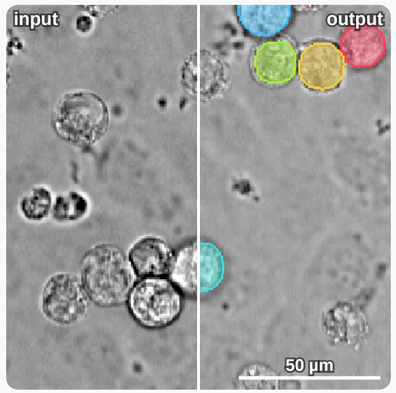
A Deep-learning examples

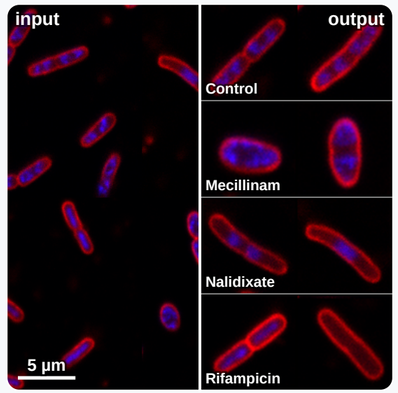
Turn images into measurements



Detection

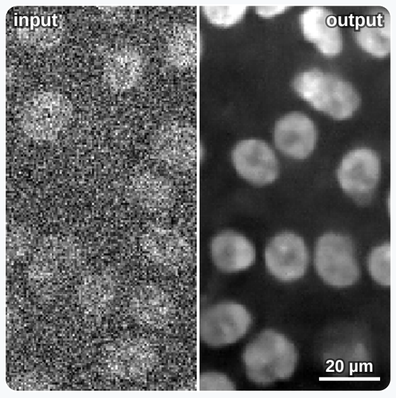


Segmentation

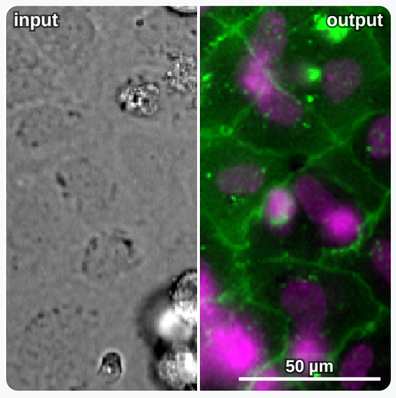


Classification

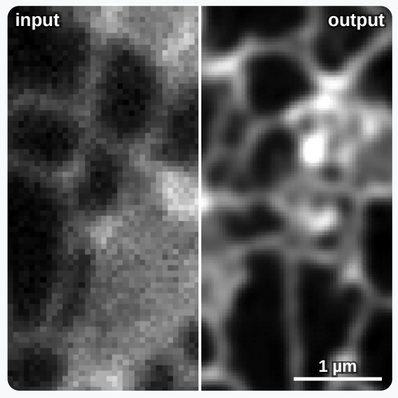
Predict image content



Denoising



Artificial labelling



Super-resolution

B Four ways to use an LLM

1 Ask for advice

software, parameters, approaches

"Which tools can segment touching nuclei in 2D fluorescence images, and which parameters matter most?"

"StarDist or Cellpose; check the probability threshold and object size."

2 Ask for code

e.g. Python, Fiji macro

"Write a Python script that segments nuclei and measures their area in µm². Do not silently resize or normalise."

```
labels = segment(img)
area_um2 = area_px * px_um**2
```

3 Let an agent run it

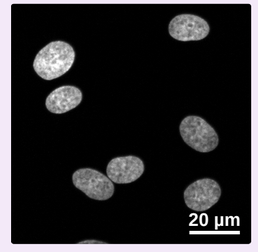
runs and optimises the workflow

"Optimise the nuclei segmentation on the training folder and report the F1. Do not open the held-out folder."

```
run 1: F1 = 0.62
run 4: F1 = 0.81, saved
```

4 Ask for the answer

multimodal model, direct answer



"How many nuclei are in this image?"

asked 3 times:

run 1: "7 nuclei"
run 2: "8 nuclei"
run 3: "7 nuclei"

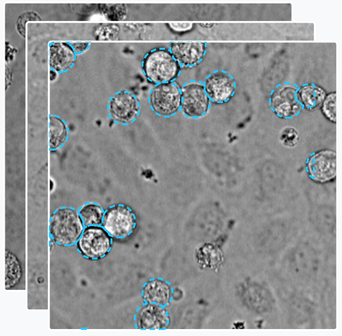
C Start an image-analysis project

1 Define the measurement

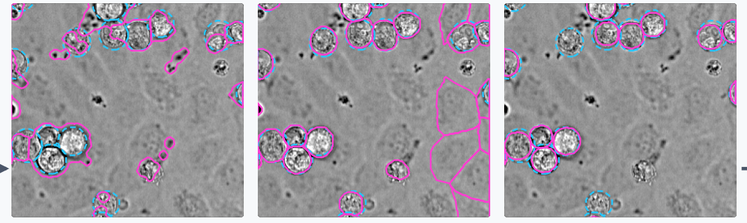
Can we segment the cancer cells in these brightfield images so that we can track them?

Measure: cell tracks

2 Build a reference test set



3 Compare candidate approaches and optimise

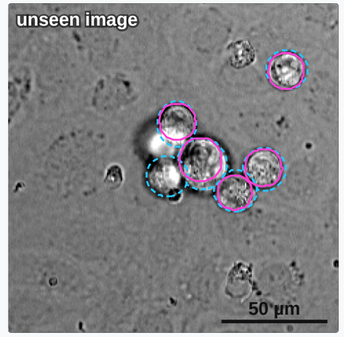


Threshold	Pre-trained generalist	Custom-trained
0.38	0.78	0.86

candidate — manual annotation

bars: F1 score

4 Final check on fresh images



5 Track the cells

